Data Management and Databases

Welcome to the Data Management and Databases course! 👋

Introduction to databases

- The learning objectives for this week are:
  - Knowing what kind of topics are covered during the course
  - Knowing the course schedule and how the course is assessed
  - Knowing the objective of **data management**
  - Knowing the meaning of the term **database** and **database management system** (DBMS)
  - Knowing what kind of functions a full-scale DBMS should provide
  - Knowing and advantages of the **database approach** over the **file-based approach** in data management

About the course

- During the course, we will learn among other things:
  - The key concepts and terminology of data management and databases
  - Design and document the database's structure based on the requirements
  - Retrieve and manipulate the database's data with SQL
- During the weekly sessions we dive into new topics and reinforce learning through hands-on tasks
- The teaching session schedule can be found on the Moodle's "Course outline" page
- There are weekly exercises that need to be submitted in Moodle **before the next week's session**

Schedule

- The course takes place in **two teaching periods**
- During the first period we will some basic concepts of databases and how to operate databases using the SQL query language
- The first period is followed by the **mid-term exam**, which only covers SQL
- During the second period we will mostly cover database design and a few SQL related topics
- Throughout the second period we will also be working on a **case assignment** group work related to database design
- The second period is followed by the **final exam**, which covers everything except SQL

Assessment

- ⚠️ To confirm the course participation the following need to be submitted **before the second week's session**:
  - The first part of your learning diary
  - The first week's exercises (orientation exercises and introduction exercises)
- The course assessment is based on the combined points from two exams:
  - The **mid-term exam**, half way through the course, will cover SQL operations
  - The **final exam**, at the end of the course, will cover rest of the course topics
- Passing grade has the following requirements:
  - At least 40% points in each exam and 50% exam point average from the two exams
  - At least 60% of the exercises, case assignment and learning diary must be submitted **before the end of the final course week**

Database

> *"A representation of facts or ideas in a formalized manner capable of being communicated or manipulated by some process."*
> — Definition for the word *"Data"* in Oxford Languages

- In today's digital world, we constantly access and manipulate stored information. For example, when we open an messaging application, we can view previously sent messages and send new ones
- Individual pieces of information, such as a user's email address or a message's content, are called **data**. An organized collection of related data forms a **database**
- A database is the backbone of most information systems. It enables large amounts of data to be stored efficiently while supporting fast retrieval and modification of that data
- There are many different database technologies, such as SQL Server, PostgreSQL, and MongoDB

Definition of database

- Due to changing requirements, database systems have evolved significantly over the years, and so has the definition of the term *"database"*:

> *"A database consists of some collection of persistent data that is used by the application systems of some given enterprise."*
> — Date in 1990

> *"A database is a shared collection of logically related data and a description of this data, designed to meet the information needs of an organization."*
> — Connolly & Begg in 2005

Definition of database

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- *"persistent data"*: data is commonly in **permanent storage** and doesn't vanish, when e.g. the server machine shuts down
- *"shared collection"*: database is **accessible** to specific applications, users, and organizations
- *"logically related data"*: pieces of data in a database are **connected**, e.g. a message is connected to the sender and the receiver user
- *"description of this data"*: in addition to the actual data (such as a user's phone number), the database also contains **metadata**, including table and column names
- *"meet the information needs"*: the stored data is **use-case specific**, e.g. a messaging application needs to store information about users and messages

Data management

- **Data management** is the process of developing, maintaining, and coordinating database systems to ensure that data is stored, managed, and accessed efficiently
- A database system consists of five major components:
  - **Hardware**: the physical devices and infrastructure (such as servers, storage devices, and networking equipment) that support the database system
  - **Software**: the database management system (DBMS) and related applications used to manage and access data
  - **Data**: the collection of stored information (the database) that the system manages
  - **Procedures**: the documented policies, guidelines, and instructions that define how the database system is used, maintained and secured
  - **Users**: the people and applications that interact with the database system, including database administrators, developers, and end users

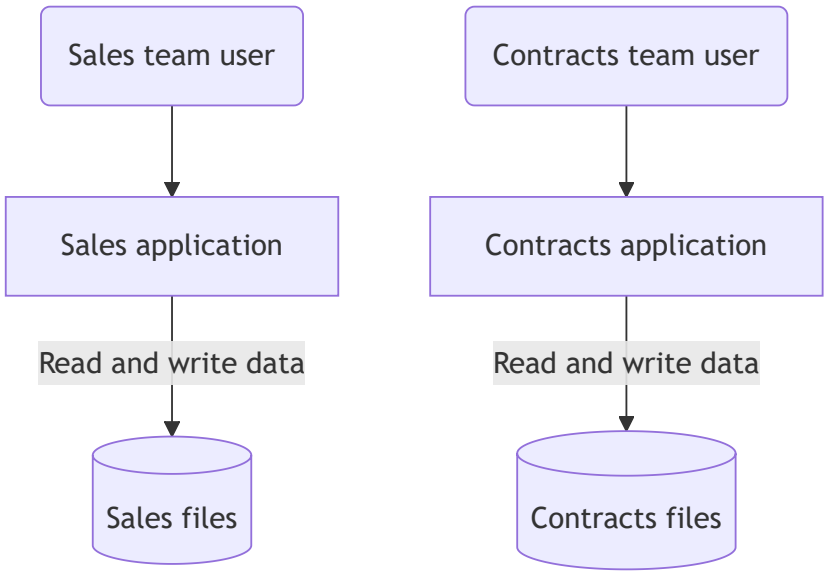
The objective of data management

- The objective of data management is to design, implement, coordinate, and maintain database systems in such a way that all the required data is:
  - Valid and consistent
  - Up to date
  - Available in the required format
  - Available when needed
  - Retrievable fast enough
  - Safe from different types of technical failures and accidents
  - Protected from unauthorized access and other types of misuse

Data management example

- Let’s consider the following information needs for a database:
  - A real estate company is renting properties
  - Each property has a property owner and a lease if the property is rented
  - Each lease has a client who is renting the property from the owner
  - The company has a **sales team** responsible for finding clients for the available properties and a **contracts team** responsible for managing the leases
- Next, we will have a look at two possible approaches of managing this data, which are the **file-based** and **database management system** approaches

File-based approach in data management



- Sales and contracts teams have data relevant for their needs in **separate files** (e.g. spreadsheets)
- Each team uses a separate application that defines and manages data in application-specific files
- Each file has a specific format and the application that uses a file depends on knowledge about its format

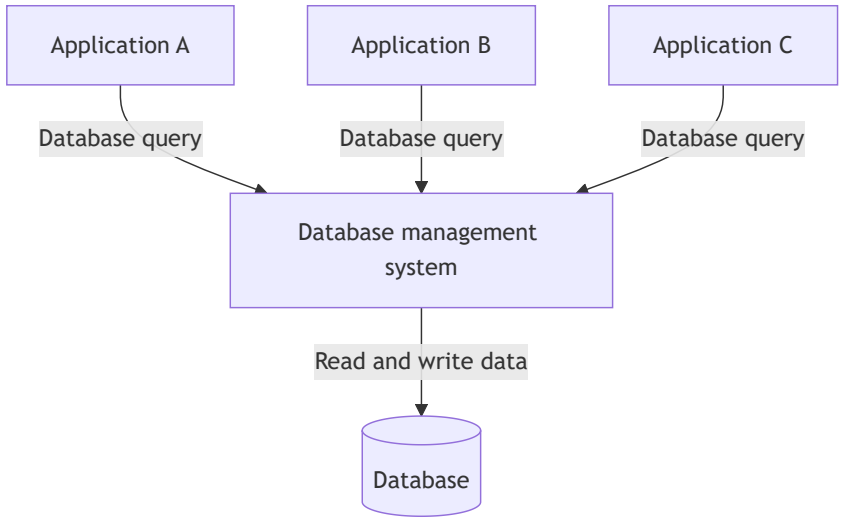
File-based approach in data management

- The sales application operates with the following **sales files**:
  - *property-owners* (ownernumber, firstname, lastname, address, phone)
  - ⚠️ *properties* (propertynumber, ownernumber, street, city, postcode, size, roomcount, rent)
  - ⚠️ *clients* (clientnumber, firstname, lastname, address, phone, maxrent, sizepreference)
- And the contracts application with the separate **contracts files**:
  - *leases* (leasenumber, propertynumber, clientnumber, startdate, enddate, deposit)
  - ⚠️ *properties* (propertynumber, street, city, postcode, rent)
  - ⚠️ *clients* (clientnumber, firstname, lastname, address, phone)
- The same client and property information is stored in **both sales and contracts files**. Such **data reduncancy** will lead to inconsistent information, e.g. the same client having a different phone number in different files

Problems of file-based approach

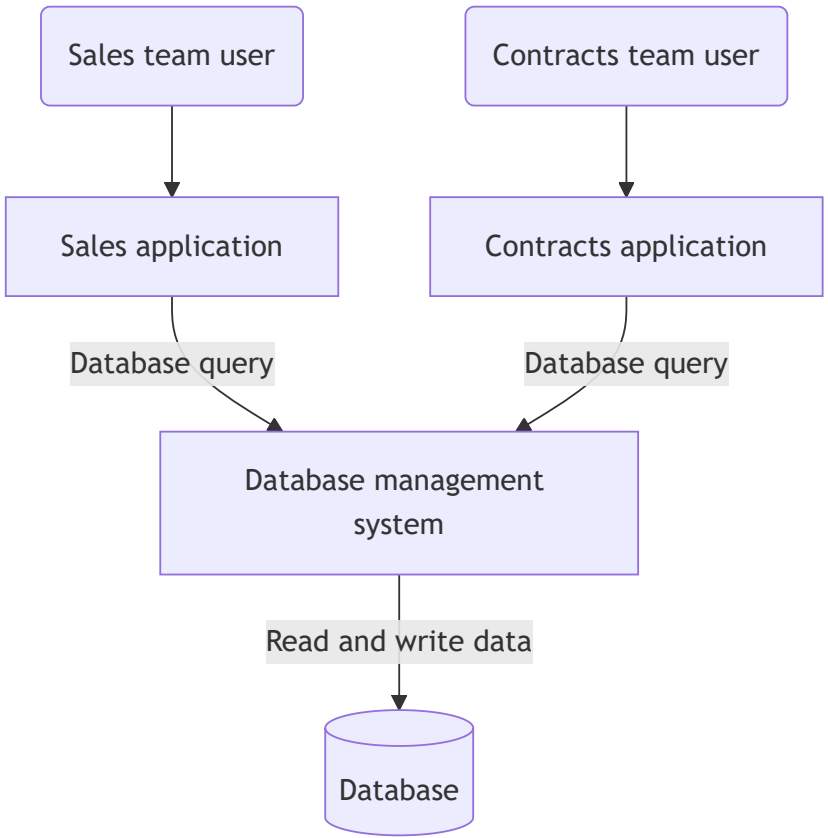
- **Data dependence**: code is being tightly coupled with the file structure and if it is modified, all applications that use the file have to be changed accordingly
- **Data redundancy**: the same information is stored in multiple files (e.g. the client files in both sales and contacts teams), which will lead to inconsistency
- **Difficulty in accessing data**: applications are written to satisfy particular functions and any new requirement needs a new application or changes in an existing one
- **No provision for security and shared access to the data**: there is no service for providing user access to some, but not all, data
- **Lack of coordination and standardisation**: there is no centralised control of enterprise data, which makes it difficult to enforce standards and share data structures between applications

Database Management System (DBMS)



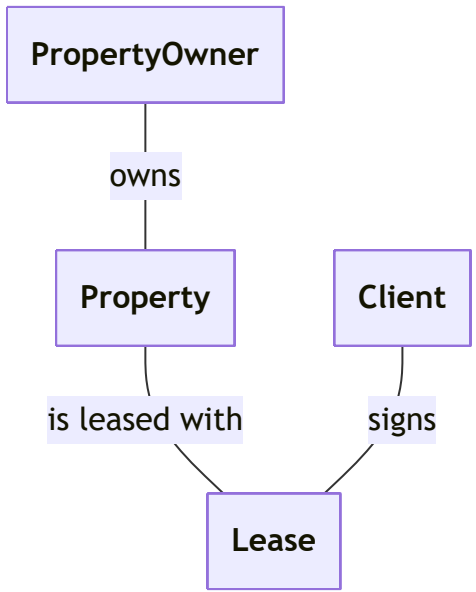
- The limitations of the file-based approach can be overcome by using a separate software system called a **database management system** (DBMS), which manages and controls all access to the database
- DBMS allows users to define the structure of a database using a a **data definition language** (DDL)
- DBMS also enables users to insert, update, delete, and retrieve data from the database using a **data manipulation language** (DML)

DBMS approach in data management



- Data for both sales and contracts team is in the **same database**, without data redundancies
- Each team uses a separate application that communicates with the **same DBMS** using a data manipulation language
- The DBMS retrieves and manipulates data in the database on behalf of the application
- The key difference is that **the applications don't access the data directly**, but trough the DBMS

DBMS approach in data management



- In the database, the sales and contracts details can be structured into the following **database tables**:
  - `PropertyOwner` (ownernumber, firstname, lastname, address, phone)
  - `Property` (propertynumber, ownernumber, street, city, postcode, size, roomcount, rent)
  - `Client` (clientnumber, firstname, lastname, address, phone, maxrent, sizepreference)
  - `Lease` (leasenumber, propertynumber, clientnumber, startdate, enddate, deposit)
- Now, there's a **common structure** for the organization's data without duplication and structural differences

Advantages of the DBMS approach

- **Data independence**: applications don't need to know about the physical level storage structures, which improves data accessibility, maintainability of the applications and reusability of existing data
- **Effective access to data**: multiple users can access the data concurrently using a standard database language with both programmatic and interactive interfaces
- **Data integrity**: integrity can be maintained by using user-defined integrity constraints
- **Data security**: security restrictions can be applied on detailed level
- **Coordination and standardisation**: centralised data administration eliminates redundancy and enforces standards
- **Increased application development productivity**: standard database language simplifies the development of applications and provides portability

Functions of a DBMS

- The most fundamental function of DBMS is **data storage and data retrieval and update operations**
- This function should be provided in such a way that the physical level storage structures are completely hidden from the user, which enables data independence
- Other important functions of a DBMS are:
  - Integrity services
  - Transaction support
  - Concurrency control services
  - Recovery services
  - Authorization services
  - User-accessible system catalog
  - Support for data communication

Functions of a DBMS

- **Integrity services**: protects data integrity (correctness and consistency of stored data) based on user-specifiable **integrity constraints** (consistency rules)
- **Transaction support**: protects database integrity by providing reliable units of work (database transactions) that allow correct recovery from failures and providing isolation between users accessing the database concurrently
- **Concurrency control services**: allows shared access to of the database and ensure that the database is updated correctly when multiple users are updating the database concurrently
- **Recovery Services** database is able to recover to a consistent state after a system crash, media failure, a hardware or software error
- **Authorization services**: controls access to the data and ensure that the data is secure
- **User-accessible system catalog**: describes the database itself
- **Support for data communication**: the DBMS should be accessible remotely over network

Relational database managament systems (RDBMS)

- **Relational database management systems** (RDBMS) are among the most widely used types of database management systems
- The term *"relational"* refers to the way data is organized into related tables
- **Structured Query Language** (SQL) is the standard language for querying and managing data in RDBMSs
- SQL includes both **data definition language** (DDL) for defining database structures and **data manipulation language** (DML) for querying and modifying data
- Popular RDBMS products include SQL Server, MySQL, and PostgreSQL. In this course, we will focus on the SQL Server RDBMS

```
-- Example of DML syntax in SQL
SELECT clientnumber, firstname, lastname, address FROM Client
WHERE maxrent < 1000
```

Summary

- **Database** is a shared collection of logically related persistent data, which is designed to meet specific information needs
- **Data management** is the development, maintenance and coordination of **database systems**
- **Database management system** (DBMS) is a software that allows users to insert, update, delete, and retrieve data from the database
- **Data independence**, **effective access to data** and **data integrity** are some of the key benefits of a DBMS approach in data management
- The most fundamental function of DBMS is **data storage and data retrieval and update operations**
- **Relational database management system** (RDBMS) are among the most popular database management systems